

RESEARCH INTERESTS

Spatiotemporal machine learning for sustainability systems; remote sensing; climate risk; wildfire spread modeling; environmental monitoring and decision support.

EDUCATION

McGill University & Mila – Quebec AI Institute <i>M.Sc. in Computer Science (GPA: 4.0/4.0)</i>	Montréal, QC Aug. 2025 – Present
Duke University / Duke Kunshan University <i>B.S. in Data Science</i>	Durham, NC / Suzhou, China Aug. 2021 – May 2025
- Thesis: <i>Monitoring Spartina Alterniflora Using Self-Supervised Learning</i>. - Study Abroad: Universidad Carlos III de Madrid (Spring 2022).	

GRANTS & AWARDS

- McGill CS Graduate Award — \$30,000/year (2025)
- McGill CS Graduate Registration Bonus — \$5,000 (2025)
- DKU Student Experiential Learning Fellowship — \$1,000 (2024)
- DKU Signature Work Research Grant — \$1,500 (2023)
- DKU Innovation Incubator — \$3,000 (2023)

RESEARCH EXPERIENCE

Machine Learning Research Intern <i>University Mohammed VI Polytechnic</i>	Jul. 2025 – Present Benquerir, Morocco
- Integrated a 10-year global crop-yield dataset with fertilizer inputs (N/P/K) across 150+ countries for downstream modeling and analysis. - Developed a transformer-based field recommendation model and an LLM+retrieval assistant (RAG) for agronomic decision support; optimized retrieval to ~2s query latency.	
Undergraduate Research Assistant <i>Duke Kunshan University</i>	Jun. 2024 – Aug. 2025 Suzhou, China
- Built a 300k-image, ground-truthed satellite dataset across 50/54 African countries; trained a DINO (ViT) model to 97% accuracy (<i>supervised by Dr. Ka Leung Lam</i>). <i>SpeedTransformer</i>: achieved 96% Top-1 accuracy on Geolife transport-mode prediction (+3% over baseline); built <i>CarbonClever</i>, a prototype app translating mobility insights into personal CO₂ reduction actions (<i>supervised by Dr. Charles Chang</i>).	
Undergraduate Research Assistant <i>University of Missouri</i>	Jan. 2025 – Jun. 2025 Columbia, MO
Developed a Perceiver IO-based model and a VAE simulator for wildfire spread using satellite and climate data; evaluated on the WSTS+ benchmark and enabled scenario-driven analyses (<i>supervised by Dr. Jordan Malof</i>).	
Undergraduate Research Fellow <i>Duke University</i>	Jun. 2023 – Aug. 2024 Durham, NC
- Mapped invasive salt-marsh species from Sentinel-2/Landsat using ViTs; delivered landscape-scale outputs presented at Duke Climate+ 2024 to inform coastal conservation policy (<i>supervised by Dr. Wenhong Li & Dr. Ding Ma</i>). - Built a geospatial database on saltwater intrusion & sea-level rise (North American Coastal Plain); mined 1,000+ papers using BERT-based topic modeling; produced an interactive ArcGIS dashboard (<i>supervised by Dr. Emily Bernhardt</i>).	

PROFESSIONAL EXPERIENCE

Data Analyst Intern

Oct. 2022 – Nov. 2022

Atos Morocco

Rabat, Morocco

- Built real-time HR dashboards used by leadership; scaled deployment across African branches to standardize reporting and decision support.

Data Analyst Intern

Jul. 2022 – Aug. 2022

XPerlean

Saint-Quentin, France

- Applied Faster R-CNN and YOLO for ceramic defect detection, improving accuracy 70%→85% and increasing production speed by 10% while reducing quality-control costs.

Data Analyst Intern

Oct. 2021 – Dec. 2021

Al Jazeera Media Institute

Doha, Qatar

- Collected and processed 200k+ social media records via APIs to analyze user engagement and behavior for internal reporting.

PUBLICATIONS & PREPRINTS

1. **O. Echchabi**, et al. *Tracking Progress Towards SDG 6 Using Satellite Imagery*. arXiv:2411.19093. [Paper](#).
2. Z. Yuandong*, **O. Echchabi***, et al. *SpeedTransformer: Predicting Human Mobility with Dense GPS and Transformers*. Under review. [Code](#).
3. **O. Echchabi**, N. Talty. *Football AI Tracker: End-to-End Player Tracking from Low-Quality Recording*. [Report](#).

*Equal contribution.

SELECTED PRESENTATIONS

- [Monitoring *Spartina alterniflora* Using Self-Supervised Learning](#). **Duke Climate+ Symposium** (2024).
- [Assessing Climate Change Risk of Rural Coastal Plains](#). **Duke Climate+ Symposium** (2023).

TEACHING EXPERIENCE

- **McGill University** — Teaching Assistant, COMP 303 (2025–Present)
- **Duke Kunshan University** — Math & CS Teaching Assistant (multiple terms)
- **FADI Academy** — Partner & Math Tutor

SERVICE & LEADERSHIP

- Mila Lab Representative; Muslims in Tech (Board Member)
- Duke Kunshan University — Resident Assistant; Student Athlete; Soccer Varsity Co-Captain; CS Club Software Developer ([website](#))
- FIFA World Cup 2022 — Team Leader, Spectator Services Volunteer

TECHNICAL SKILLS

- **Programming:** Python, Java, C/C++, SQL (Postgres), JavaScript, R
- **ML/Data:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Matplotlib
- **Systems/Tools:** Git, Docker, Google Cloud Platform, Linux
- **Web/Backend:** React, Node.js, Flask, FastAPI

LANGUAGES

- Arabic (native); French (fluent); English (fluent); Chinese (intermediate); Spanish (conversational)